

Jayant Bodkurwar

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Skills

Languages: Python, Java, SQL

Frameworks: FastAPI, Flask, Django, SpringBoot, ReactJS, Celery, RabbitMQ, Redis, Kafka, RabbitMQ

Technologies & Tools: AWS (ECS, EKS, ECR, S3, Lambda, Athena), Docker, Kubernetes, PostgreSQL, MySQL, MongoDB, Elasticsearch, Opensearch, CircleCI, GitHub Actions, Vault

Software Development & Architecture: Microservices, System Design, Performance Optimization, Scalability, Security, Unit & Integration Testing (pytest, unittest), Technical Documentation.

Education

University of Mumbai

Aug 2016 - Jun 2020

B.E. in Information Technology

Relevant Coursework: Object Oriented Programming, Databases, Data Structures and Algorithms, Operating Systems, Computer Networks, Advanced Data Structures and Algorithms

Work Experience

Serrala, Pune

Oct 2024 - Present

Senior Software Engineer

- **Core AR Automation Platform — Backend Engineering:** Engineered core backend of a multi-tenant accounts-receivable automation platform (Flask/Celery/MySQL) — 900+ commits, 116K+ lines shipped, 550+ REST endpoints, and 920+ database migrations across a production system serving live client tenants.
- **ERP & Financial-Systems Integration:** Built and maintained integrations with 8 external ERP, banking, and payment systems — NetSuite, SAP (BTP & Public Cloud), Sage Intacct, QuickBooks (Online & Desktop), Stripe, and TempWorks — automating cash application and invoice/payment sync at scale.
- **Credit Management Platform:** Delivered an end-to-end B2B credit management platform in 2 months — credit limits, applications, RBAC-driven approvals, D&B risk assessment, and daily automated scoring across 7 weighted credit KPIs — now live across 2 enterprise clients, serving credit decisions for 500K+ customers via a 328-endpoint FastAPI microservice.
- **Background Job Infrastructure:** Built and operate a Celery + RedBeat distributed background-job system processing 45,000 production jobs/day at 0% infra-failure rate — with Slack-integrated failure alerting, a custom monitoring dashboard for real-time job/worker visibility, and automatic worker autoscaling.
- **AI Agent Platform:** Built an AI-powered agent platform for accounts-receivable operations, live across 5 enterprise clients — 12 specialized LLM agents (LangChain + LangGraph, Azure OpenAI GPT-4o) spanning credit scoring, email collections, auto-drafted responses, and thread classification, plus 8 conversational sub-agents (invoice, cash application, customer, remote-deposit OCR, analytics, navigation) powering a vector-grounded RAG chatbot over Azure Cognitive Search and Cosmos DB — backed by a custom agent-observability layer and automated eval harness for accuracy scoring.
- **Optimized Database Performance for Scalability:** Led performance optimization initiatives, implementing MySQL and PostgreSQL partitioning, improving query response times by 60% and boosting system scalability.
- **Tech Stack:** Python 3.10, Celery, Redis, PostgreSQL, MySQL, RabbitMQ, AWS (S3, Lambda, ECS), Docker, Kubernetes, Microservices, CI/CD (GitHub Actions, Jenkins)

Software Developer

- **Microservices Architecture:** Designed and implemented a Spring Cloud Config Server microservice using GitHub and Vault, dynamically managing configurations for 110 microservices. Reduced ECR costs by 80% and eliminated the need for embedded properties in Docker images.
- **Cross-Platform Integration:** Engineered Python and Node.js libraries to integrate Spring Cloud Config with Django, Node.js, Play, React, and Angular, unifying configuration management across diverse microservice architectures.
- **Scalability & Storage Optimization:** Developed and deployed a Django-based File Upload Service, integrated with AWS S3, EFS, EKS, and Azure Storage, successfully managing 8M+ file uploads per month with a 99.8% success rate.
- **CI/CD Pipeline Transformation:** Led the migration to a Tekton-based CI/CD pipeline, enabling single-image deployment across 110 microservices. Enhanced deployment efficiency, reducing errors by 60%.
- **Cloud Cost Optimization:** Achieved a 40% reduction in AWS costs by optimizing AWS GuardDuty, S3, ELK pipeline, VPC, and CloudWatch. Built a billing dashboard using AWS Athena, aggregating cost data from 10 AWS accounts and implementing an automated alert system for finance teams to monitor cost spikes.
- **Tech Stack:** Python, Django, JavaScript, Node.js, Java, Spring Boot, PostgreSQL, MongoDB, Kubernetes, Docker, AWS (EKS, S3, Athena, EFS), HashiCorp Vault, Ansible, Tekton

LTIMindtree, Pune, India

Jan 2020 - Jan 2022

Senior Product Engineer

- **Kubernetes Migration & Automation:** Led the migration of 54 microservices to Kubernetes, designing and deploying ConfigMaps, Deployments, Services, Ingress, and PersistentVolumes. Developed a one-click deployment system that reduced product setup time by **99%**, transforming release efficiency.
- **Microservice Development with Flask:** Designed and implemented a Flask-based microservice to dynamically provision and start Jupyter, RStudio, Spark, and SciPy notebooks on demand. Enabled seamless resource selection for users, improving system scalability and flexibility.
- **External Data Integration:** Built a scalable data access module that streamlined external data ingestion into notebooks, enhancing analysis capabilities. Integrated support for MySQL, PostgreSQL, NAS, AWS S3, Azure Blob, and Google Cloud Storage, ensuring seamless multi-cloud compatibility.
- **Containerization & Cloud Infrastructure:** Containerized all services using Docker, optimized for Kubernetes (EKS), and deployed via Helm. Improved resource utilization and deployment consistency across environments.
- **Security & Secrets Management:** Implemented HashiCorp Vault for secure credential and secret management across all services, enhancing platform security and access control.
- **Tech Stack:** Python, Flask, Django, MySQL, PostgreSQL, Kubernetes, Docker, AWS EKS, S3, EFS, NAS, HashiCorp Vault, Helm

Projects

Scalable and Distributed Food Ordering System

Designed and developed a microservices-based food ordering platform with asynchronous messaging and real-time tracking.

- **Microservices Architecture:** Implemented independent services for order processing, vendor management, and communication, enabling efficient scaling and fault tolerance.
- **Asynchronous Messaging with RabbitMQ:** Used a fanout exchange pattern to distribute order messages across multiple services, ensuring high availability and responsiveness.
- **Dual Database Strategy:** Utilized PostgreSQL for managing menu data and MySQL for order processing, showcasing advanced database segregation techniques.
- **Order Processing:** Designed an order service that publishes messages to RabbitMQ, allowing seamless order tracking and automated status updates.
- **Real-Time Communication:** Built an email service using Celery workers to send order confirmations, ensuring timely notifications to customers.
- **Scalability and Fault Tolerance:** Leveraged Docker and Kubernetes for containerization and deployment, enabling horizontal scaling of individual services.

GitHub Repository: github.com/jayantreddy181198/delivery-system

Awards and Certificates

- Meta Front-End Developer Professional Certificate on Coursera